

Python Encapsulation Practice

30 Questions — Easy English

Topics: Public Variables, Protected Variables, Private Variables

How to practice: Read one question, write the code yourself, run it, and check the output. The questions become harder slowly.

1. Public Variables — 10 Questions

Question 1: Create a class **Student**. Add a public variable **name**. Create one object and print the name.

Question 2: Create a class **Car**. Add a public variable **color**. Create an object and print the color.

Question 3: Create a class **Book**. Add two public variables: **title** and **price**. Create an object and print both.

Question 4: Create a class **Employee**. Add a public variable **salary**. Create an object and change the salary from outside the class.

Question 5: Create a class **Student** with public variables **name** and **marks**. Create an object and print both values.

Question 6: Create a class **Product** with a public variable **price**. Change the price using the object and print the new price.

Question 7: Create a class **Person** with a public variable **age**. Create two objects with different ages. Print both ages.

Question 8: Create a class **Student** with a public variable **marks** as a list. Add [80, 90, 70] and print the list.

Question 9: Create a class **BankAccount** with a public variable **balance**. Add 500 to the balance from outside the class and print it.

Question 10: Interview question: A public variable can be changed from outside the class. Write a small program to show this.

2. Protected Variables — 10 Questions

Question 1: Create a class **Student**. Add a protected variable `_name`. Create an object and print the name.

Question 2: Create a class **Employee** with a protected variable `_salary`. Create an object and print the salary.

Question 3: Create a class **Car** with a protected variable `_speed`. Create a method to print the speed.

Question 4: Create a class **Person** with `_age`. Create a child class **Student** and print `_age` from the child class.

Question 5: Create a parent class **Animal** with `_sound`. Create a child class **Dog** and use `_sound` inside Dog.

Question 6: Create a class **BankAccount** with `_balance`. Add a deposit method that changes `_balance`.

Question 7: Create a class **Game** with `_score`. Create a child class and increase the score from the child class.

Question 8: Create a class **Employee** with `_salary`. Create a Manager class that reads the salary and prints it.

Question 9: Try to access a protected variable from outside the class. What happens? Write code and see the result.

Question 10: Interview question: Does Python completely stop you from accessing a protected variable? Write a simple example to check.

3. Private Variables — 10 Questions

Question 1: Create a class **Student**. Add a private variable `__name`. Create a method to print the name.

Question 2: Create a class **Employee** with a private variable `__salary`. Create a method called `show_salary()`.

Question 3: Create a class **BankAccount** with private variable `__balance`. Create a deposit method.

Question 4: Create a class **User** with private variable `__password`. Create a method to check the password.

Question 5: Create a class **Counter** with private variable `__count`. Create an `increment()` method.

Question 6: Create a class **Student** with private variable `__marks`. Create a method that returns the marks.

Question 7: Try to print a private variable directly from outside the class. What error do you get?

Question 8: Create a class **BankAccount** with `__balance`. Add `deposit()` and `withdraw()` methods. Do not allow the balance to become negative.

Question 9: Create a class with `__name`. Print the object's `__dict__`. Find how Python changes the private variable's name.

Question 10: Interview question: Create a parent class with `__salary` and a child class. Try to use `__salary` inside the child. Explain what happens.